

Numerical Linear Algebra First-Year Exam, May 2025

Do 4 (four) problems.

1. Assume $A \in \mathbb{C}^{m \times m}$.

- (a) Show that A has a Schur decomposition.
- (b) If A is normal, show that A is diagonalizable.

2. Suppose A is Hermitian positive definite.

- (a) Prove that each principal submatrix of A is Hermitian positive definite.
- (b) Prove that an element of A with largest magnitude lies on the diagonal.
- (c) Prove that A has a Cholesky decomposition.

3. This problem has two parts.

- (a) Show that $\|x\|_\infty$ is equivalent to $\|x\|_2$ for all $x \in \mathbb{R}^n$. That is to find C and c such that $c\|x\|_\infty \leq \|x\|_2 \leq C\|x\|_\infty$, for all $x \in \mathbb{R}^n$. Note that the constants should be determined so that the equalities hold for some nonzero $x \in \mathbb{R}^n$.
- (b) Show that $\|QA\|_2 = \|A\|_2$ if Q is a unitary matrix.

4. Assume that $A \in \mathbb{C}^{n \times n}$ and there exists $p \geq 1$ such that $\|A\|_p < 1$, where $\|\cdot\|_p$ is a vector-induced matrix norm.

- (a) Prove that $I - A$ is invertible.
- (b) Prove that

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k.$$

- (c) Prove that $\|A\|_q \|A^{-1}\|_q \geq 1$, $\forall 1 \leq q \leq \infty$.
- (d) Prove that

$$\frac{1}{1 + \|A\|_p} \leq \|(I - A)^{-1}\|_p \leq \frac{1}{1 - \|A\|_p}.$$

5. Let $A = U\Sigma V^*$ be the singular value decomposition of $A \in \mathbb{C}^{m \times n}$. Let u_j denote column j of U .

- (a) Suppose $\text{rank}(A) = p < n < m$. Show $\{u_1, u_2, \dots, u_p\}$ is a basis for $\text{Col}(A)$ and $\{u_{p+1}, u_{p+2}, \dots, u_m\}$ is a basis for $\text{Null}(A^*)$.
- (b) Suppose A is full rank and $x \neq 0$. Let σ_i , $i = 1, \dots, n$, be the singular values of A . Show

$$\sigma_1 \geq \frac{\|Ax\|_2}{\|x\|_2} \geq \sigma_n > 0.$$

If you want to use the property that $\|A\|_2 = \sigma_1$, then you must prove that it holds.